

Assignment2b

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Loading Datasets

```
NOAAnew <- read.csv('NOAAnew.csv')
NOAAold <- read.csv("NOAA+GISS.csv")
set.seed(123)

#renaming columns so that both files have same column names
names(NOAAnew)[names(NOAAnew) == "X.disaster"] <- "disasters"
names(NOAAold)[names(NOAAold) == "X.disaster"] <- "disasters"
names(NOAAold)[names(NOAAold) == "delta temp"] <- "delta.temp"

head(NOAAnew)
```

```
##   year disasters delta.temp
## 1 1980         3      0.27
## 2 1981         1      0.33
## 3 1982         3      0.13
## 4 1983         5      0.30
## 5 1984         2      0.16
## 6 1985         5      0.12
```

```
head(NOAAold)
```

```
##   year disasters delta.temp
## 1 1980         3      0.27
## 2 1981         1      0.33
## 3 1982         3      0.13
## 4 1983         5      0.30
## 5 1984         2      0.16
## 6 1985         5      0.12
```

Comparing Bends

```
#function to create and fit spline
bends <- function(data_new , data_old , x , y , xLabel , yLabel , title)
{
  if(x == "temperature")
  {
    #smooth spline for temp vs disasters with df=7
    fit_new <- smooth.spline(data_new[[x]] , data_new[[y]] , df = 7)
    fit_old <- smooth.spline(data_old[[x]] , data_old[[y]] , df = 7)
  }
  else
```

```

{
  #smooth spline for year vs disaster with df=7
  fit_new <- smooth.spline(data_new[[x]] , data_new[[y]] , df = 7)
  fit_old <- smooth.spline(data_old[[x]] , data_old[[y]] , df = 7)
}

#creating data of predictions
x_range_new <- seq(min(data_new[[x]]), max(data_new[[x]]), length.out = 100)
x_range_old <- seq(min(data_old[[x]]), max(data_old[[x]]), length.out = 100)
pred_new <- predict(fit_new, x_range_new)
pred_old <- predict(fit_old, x_range_old)

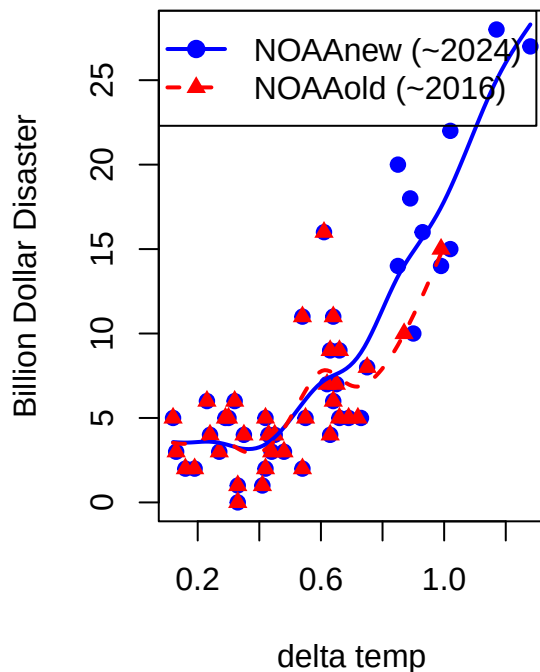
#plotting the curves
plot(data_new[[x]] , data_new[[y]] , col = "blue" , pch = 19 ,
      xlim = range(c(data_new[[x]] , data_old[[x]])) , ylim = range(c(data_new[[y]] , data_old[[y]])) ,
      xlab = xLabel , ylab = yLabel , main = title)
points(data_old[[x]] , data_old[[y]] , col = "red" , pch = 17)
lines(pred_new$x , pred_new$y , col = "blue" , lwd = 2)
lines(pred_old$x , pred_old$y , col = "red" , lwd = 2 , lty = 2)
legend("topleft" , legend = c("NOAAnew (~2024)" , "NOAAold (~2016)") , col = c("blue" , "red") , pch = c(19 , 17))

return(list(fit_new = fit_new , fit_old = fit_old))
}

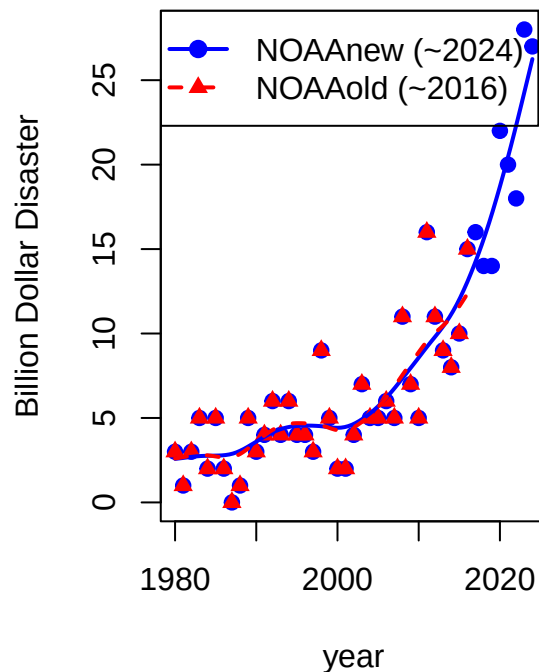
par(mfrow = c(1 , 2))
fit_temp <- bends(NOAAnew , NOAAold , "delta.temp" , "disasters" , "delta temp" , "Billion Dollar Disaster")
fit_year <- bends(NOAAnew , NOAAold , "year" , "disasters" , "year" , "Billion Dollar Disaster")

```

Temperature vs Disaster



Year vs Disaster



Bootstrap

```
boot_spline <- function(x , y , x_new , nboot = 10000 , df = NULL , conf.level = .90)
{
  n <- length(x)

  #original model fit
  if (is.null(df))
  {
    fit_orig <- smooth.spline(x , y)
  }
  else
  {
    fit_orig <- smooth.spline(x , y , df = df)
  }

  #getting fitted values and residuals
  fitted_orig <- predict(fit_orig , x)$y
  resid_orig <- y - fitted_orig

  #predictions for new x values
  pred_orig <- predict(fit_orig , x_new)$y

  #bootstrap matrix of predictions
  boot_preds <- matrix(NA , nrow = nboot , ncol = length(x_new))

  for (i in 1:nboot)
  {
    #resampling residuals
    boot_resid <- sample(resid_orig , n , replace = TRUE)

    #creating bootstrap y values
    boot_y <- fitted_orig + boot_resid

    #fitting spline to bootstrap data
    if (is.null(df)) {
      boot_fit <- smooth.spline(x , boot_y)
    } else {
      boot_fit <- smooth.spline(x , boot_y , df = df)
    }

    #prediction for new x values
    boot_preds[i, ] <- predict(boot_fit , x_new)$y
  }

  #prediction bounds
  alpha <- 1 - conf.level
  lower_bound <- apply(boot_preds , 2 , quantile , probs = alpha/2)
  upper_bound <- apply(boot_preds , 2 , quantile , probs = 1 - alpha/2)

  return(list(x_new = x_new , predictions = pred_orig , lower = lower_bound , upper = upper_bound , boot_preds = boot_preds))
}
```

Prediction Boudns for 2025 and 2028 (Year as Predictor)

```
new_years <- c(2025, 2028)

boot_year_results <- boot_spline(x = NOAAnew$year , y = NOAAnew$disasters , x_new = new_years , nboot

#displaying bootstrap results
cat("90% Prediction Bounds for Billion Dollar Disasters:")

## 90% Prediction Bounds for Billion Dollar Disasters:
for (i in 1:length(new_years))
{
  cat(sprintf("Year %d: %.1f [90% CI: %.1f - %.1f]\n" , new_years[i] , boot_year_results$predictions[i]
}

## Year 2025: 28.2 [90% CI: 24.9 - 30.0]
## Year 2028: 34.2 [90% CI: 28.7 - 36.7]

#plotting results with prediction bounds
plot(NOAAnew$year , NOAAnew$disasters , xlim = c(min(NOAAnew$year), 2030) , ylim = c(0, max(c(NOAAnew$d

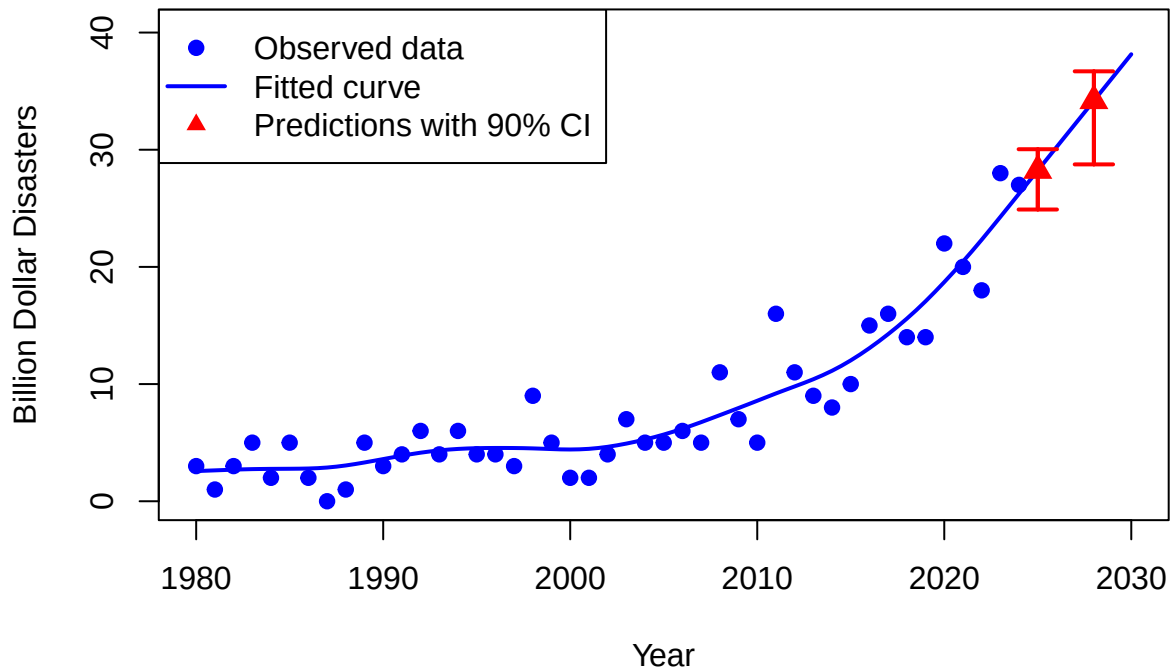
#adding fitted line on graph
year_seq <- seq(min(NOAAnew$year) , 2030 , by = 0.5)
pred_seq <- predict(boot_year_results$fit , year_seq)
lines(pred_seq$x , pred_seq$y , col = "blue" , lwd = 2)

#adding predictions and bounds on graph
points(new_years , boot_year_results$predictions , col = "red" , pch = 17 , cex = 1.5)
segments(new_years, boot_year_results$lower , new_years , boot_year_results$upper, col = "red" , lwd = 2)

# Add error bars
arrows(new_years , boot_year_results$lower , new_years , boot_year_results$upper , code = 3 , angle = 90)

legend("topleft" , legend = c("Observed data" , "Fitted curve" , "Predictions with 90% CI") , col = c("black" , "blue" , "red") ,
```

Disasters vs Year with 90% Prediction Bounds



Extraploation using Temperature

```
#model of temp over time
temp_time_fit <- smooth.spline(NOAAnew$year , NOAAnew$delta.temp , df = 5)

#prediction of temperatures for 2025 and 2028
pred_temps_2025_2028 <- predict(temp_time_fit, new_years)$y
cat(sprintf("Prediction of temperature anomaly for 2025: %.3f\n" , pred_temps_2025_2028[1]))

## Prediction of temperature anomaly for 2025: 1.173
cat(sprintf("Prediction of temperature anomaly for 2028: %.3f\n" , pred_temps_2025_2028[2]))

## Prediction of temperature anomaly for 2028: 1.291

#bootstrap prediction using temp
boot_temp_results <- boot_spline(x = NOAAnew$delta.temp , y = NOAAnew$disasters , x_new = pred_temps_2025_2028)

for (i in 1:length(new_years))
{
  cat(sprintf("Year %d (temp=%.3f): %.1f [90% CI: %.1f - %.1f]\n" , new_years[i] , pred_temps_2025_2028[i] , boot_temp_results[i,1] , boot_temp_results[i,2] , boot_temp_results[i,3]))
}

## Year 2025 (temp=1.173): 23.9 [90% CI: 20.8 - 26.7]
## Year 2028 (temp=1.291): 28.0 [90% CI: 23.9 - 32.0]
```

Significance of Bend

```
#function to compare the curvature between old and new data
bend_sginificance <- function(data_new , data_old , x_var , y_var)
```

```

{
  fit_new <- smooth.spline(data_new[[x_var]] , data_new[[y_var]])
  fit_old <- smooth.spline(data_old[[x_var]] , data_old[[y_var]])

  #second derivatives curvature at common points
  x_common <- seq(min(data_old[[x_var]]), max(data_old[[x_var]]), length.out = 50)

  #getting second derivatives
  deriv2_new <- predict(fit_new , x_common , deriv = 2)$y
  deriv2_old <- predict(fit_old , x_common , deriv = 2)$y

  #comparing average absolute curvature
  avg_curv_new <- mean(abs(deriv2_new))
  avg_curv_old <- mean(abs(deriv2_old))

  #ratio
  curv_ratio <- avg_curv_new / avg_curv_old

  return(list(avg_curv_new = avg_curv_new , avg_curv_old = avg_curv_old , ratio = curv_ratio , deriv2_n

}

#Checking the bends
bend_year <- bend_sginificance(NOAAnew , NOAAold , "year" , "disasters")
bend_temp <- bend_sginificance(NOAAnew , NOAAold , "delta.temp" , "disasters")

cat("Curvature Analysis of Year vs Disasters:")

## Curvature Analysis of Year vs Disasters:
cat(sprintf("Average curvature NOAAnew: %.3f" , bend_year$avg_curv_new))

## Average curvature NOAAnew: 5.981
cat(sprintf("Average curvature NOAAold: %.3f" , bend_year$avg_curv_old))

## Average curvature NOAAold: 0.011
cat(sprintf("Ratio new/old: %.3f" , bend_year$ratio))

## Ratio new/old: 539.896
cat("Curvature Analysis of Temperature vs Disasters:")

## Curvature Analysis of Temperature vs Disasters:
cat(sprintf("Average curvature NOAAnew: %.3f" , bend_temp$avg_curv_new))

## Average curvature NOAAnew: 34.080
cat(sprintf("Average curvature NOAAold: %.3f" , bend_temp$avg_curv_old))

## Average curvature NOAAold: 105.565
cat(sprintf("Ratio new/old: %.3f" , bend_temp$ratio))

## Ratio new/old: 0.323

```

```
cat("Interpretation:")
```

```
## Interpretation:
```

```
if (bend_year$ratio > 1.5)
{
  cat("The bend in the Year-Disasters relationship is significantly more pronounced in the new data")
} else if (bend_year$ratio > 1.1)
{
  cat("The bend in the Year-Disasters relationship is moderately more pronounced in the new data")
} else
{
  cat("The bend in the Year-Disasters relationship is similar between old and new data")
}
```

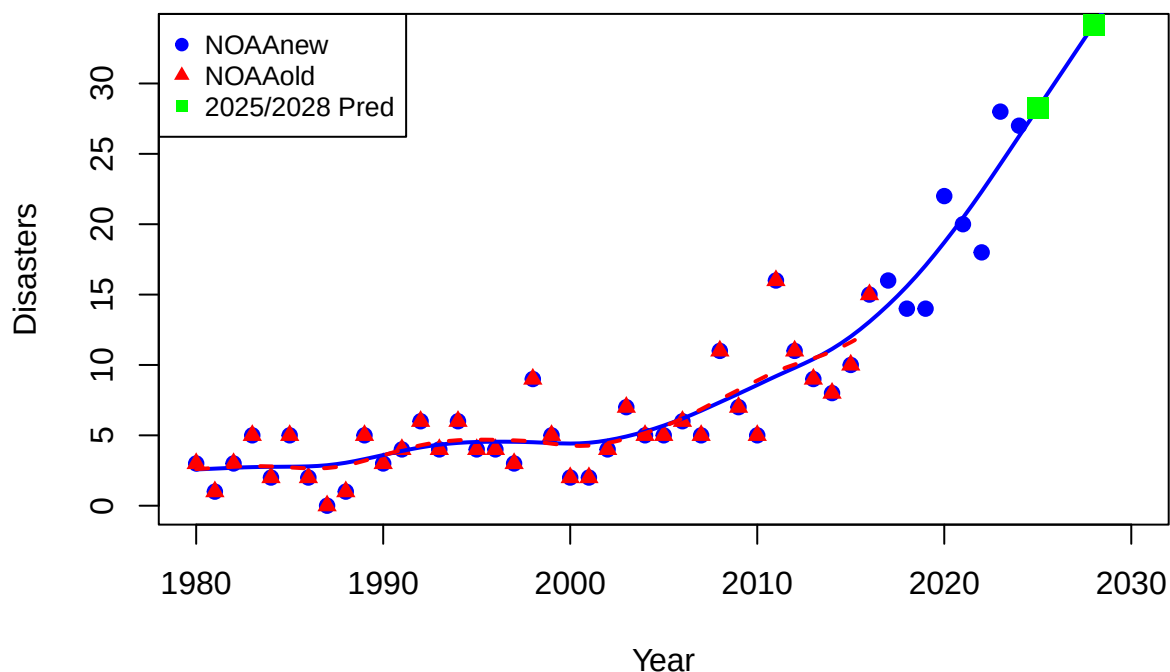
```
## The bend in the Year-Disasters relationship is significantly more pronounced in the new data
```

Visualization

```
#year vs disasters comparison
```

```
plot(NOAAnew$year , NOAAnew$disasters , col = "blue" , pch = 19 , xlim = c(1980, 2030) , ylim = c(0 , m
points(NOAAold$year , NOAAold$disasters , col = "red" , pch = 17)
year_seq_plot <- seq(1980 , 2030 , by = 0.5)
pred_new_year <- predict(smooth.spline(NOAAnew$year , NOAAnew$disasters , df = 7) , year_seq_plot)
pred_old_year <- predict(smooth.spline(NOAAold$year , NOAAold$disasters , df = 7) , seq(min(NOAAold$year
lines(pred_new_year$x , pred_new_year$y , col = "blue" , lwd = 2)
lines(pred_old_year$x , pred_old_year$y , col = "red" , lwd = 2 , lty = 2)
points(new_years , boot_year_results$predictions , col = "green" , pch = 15 , cex = 1.5)
legend("topleft" , c("NOAAnew" , "NOAAold" , "2025/2028 Pred") , col = c("blue" , "red" , "green") , pch
```

Year vs Disasters Comparison

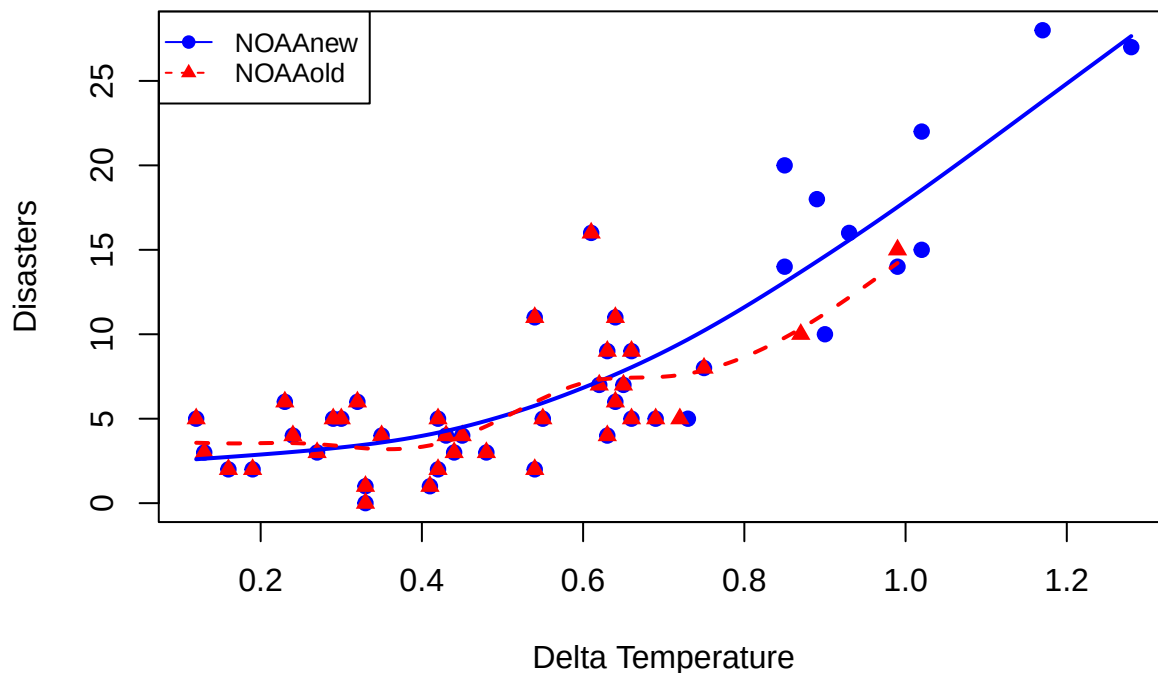


```

#temp vs disasters comparison
plot(NOAAnew$delta.temp , NOAAnew$disasters , col = "blue" , pch = 19 , xlab = "Delta Temperature" , ylab = "Disasters")
points(NOAAold$delta.temp , NOAAold$disasters , col = "red" , pch = 17)
temp_seq_new <- seq(min(NOAAnew$delta.temp) , max(NOAAnew$delta.temp) , length.out = 100)
temp_seq_old <- seq(min(NOAAold$delta.temp) , max(NOAAold$delta.temp) , length.out = 100)
pred_new_temp <- predict(smooth.spline(NOAAnew$delta.temp , NOAAnew$disasters) , temp_seq_new)
pred_old_temp <- predict(smooth.spline(NOAAold$delta.temp , NOAAold$disasters) , temp_seq_old)
lines(pred_new_temp$x, pred_new_temp$y , col = "blue" , lwd = 2)
lines(pred_old_temp$x, pred_old_temp$y , col = "red" , lwd = 2 , lty = 2)
legend("topleft" , c("NOAAnew" , "NOAAold") , col = c("blue" , "red") , pch = c(19 , 17) , lty = c(1 , 2))

```

Temperature vs Disasters Comparison

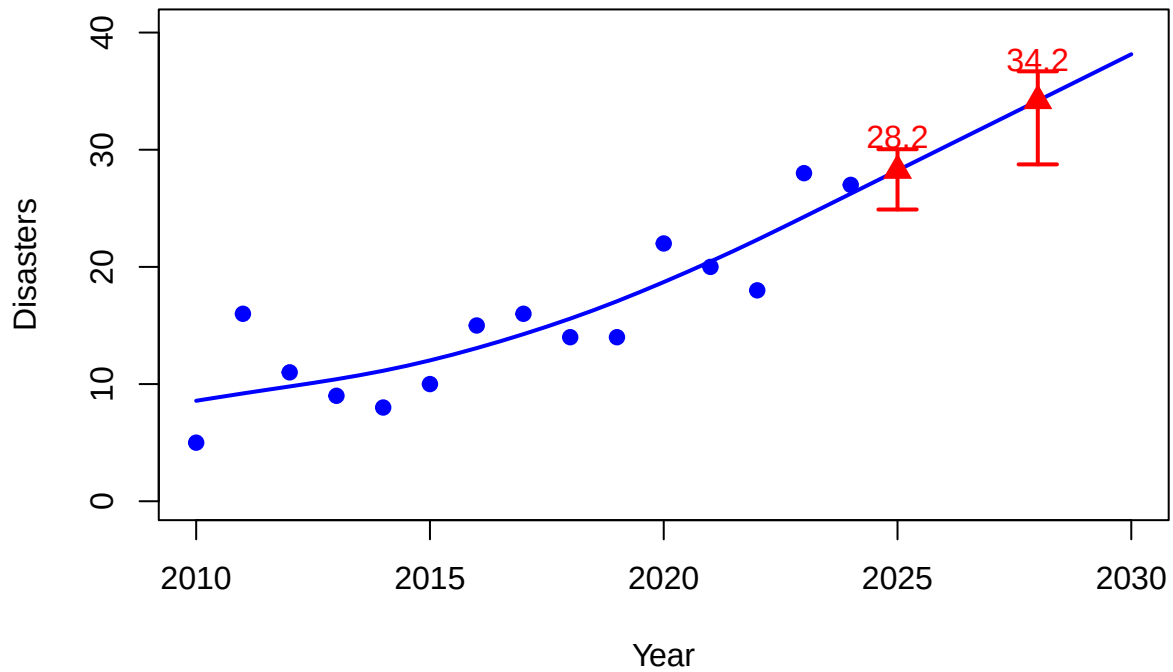


```

#prediction intervals
plot(NOAAnew$year , NOAAnew$disasters , xlim = c(2010 , 2030) , ylim = c(0 , max(boot_year_results$upper)))
year_seq_pred <- seq(2010 , 2030 , by = 0.5)
pred_curve <- predict(boot_year_results$fit , year_seq_pred)
lines(pred_curve$x , pred_curve$y , col = "blue" , lwd = 2)
points(new_years , boot_year_results$predictions , col = "red" , pch = 17 , cex = 1.5)
arrows(new_years , boot_year_results$lower , new_years , boot_year_results$upper , code = 3 , angle = 90)
text(new_years , boot_year_results$upper + 1 , labels = sprintf("%.1f" , boot_year_results$predictions))

```

Predictions with 90% CI Year-based



#curvature comparison

```
plot(bend_year$x_points , bend_year$deriv2_new , type = "l" , col = "blue" , lwd = 2 , xlab = "Year" , ylab = "Second Derivative")
lines(bend_year$x_points , bend_year$deriv2_old , col = "red" , lwd = 2 , lty = 2)
abline(h = 0 , col = "gray" , lty = 3)
legend("topright" , c("NOAAnew" , "NOAAold") , col = c("blue" , "red") , lty = c(1 , 2) , lwd = 2 , cex = 1.2)
```

Curvature Comparison of Year vs Disasters

